Name of Student: Vyankatesh Rohokale	**PRN: 26070122518**
Semester: III	**Year: AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No: 30	
TITLE: Write a JSP application to demonstrate CRUD operations.	

Problem Statement

Write a JSP application to demonstrate CRUD operations.

Learning Objectives

To implement Create, Read, Update, and Delete operations using JSP and a database.

Programme

1. Create a JSP-based application to add, update, delete, and view employee details

Index.jsp:

```
<%@ page import="java.sql.*" %>
<html>
<head>
    <title>Employee Management</title>
</head>
<body>
    <h2>Employee Management System</h2>

    <form action="save.jsp" method="post">
        Employee ID: <input type="number" name="id"
required><br><br>
        Name: <input type="text" name="name" required><br><br>
        Department: <input type="text" name="department"
required><br><br>
        Salary: <input type="number" step="0.01" name="salary"
required><br><br>
        <input type="submit" value="Add Employee">
    </form>

    <h3>Employee Records</h3>
```

```
    <table border="1" cellpadding="8">
        <tr>
            <th>ID</th>
            <th>Name</th>
            <th>Department</th>
            <th>Salary</th>
            <th>Actions</th>
        </tr>

        <%
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/EmployeeDB",
                "root",
                "admin@123"
            );

            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery("SELECT * FROM
Employee");

            while (rs.next()) {
        %>
        <tr>
            <td><%= rs.getInt("emp_id") %></td>
            <td><%= rs.getString("name") %></td>
            <td><%= rs.getString("department") %></td>
            <td><%= rs.getDouble("salary") %></td>
            <td>
                <a href="edit.jsp?id=<%= rs.getInt("emp_id")
%>">Edit</a>
                |
                <a href="delete.jsp?id=<%= rs.getInt("emp_id")
%>">Delete</a>
            </td>
        </tr>
        <%
            }
            rs.close();
            st.close();
            con.close();
        } catch (Exception e) {
            out.println("<tr><td colspan='5'>" +
e.getMessage() + "</td></tr>");
        }
        %>
    </table>
</body>
```

```
</html>
```

Edit.jsp:

```
<%@ page import="java.sql.*" %>
<%
Connection con = null;
PreparedStatement ps = null;
ResultSet rs = null;

try {
    Class.forName("com.mysql.cj.jdbc.Driver");

    con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/EmployeeDB",
        "root",
        "admin@123"
    );

    ps = con.prepareStatement("SELECT * FROM Employee WHERE
emp_id=?");
    ps.setInt(1,
Integer.parseInt(request.getParameter("id")));
    rs = ps.executeQuery();

    if (rs.next()) {
%>
<html>
<body>
    <h2>Update Employee</h2>

    <form action="update.jsp" method="post">
        Employee ID:
        <input type="number" name="id" value="<%=
rs.getInt("emp_id") %>" readonly><br><br>

        Name:
        <input type="text" name="name" value="<%=
rs.getString("name") %>" required><br><br>

        Department:
        <input type="text" name="department" value="<%=
rs.getString("department") %>" required><br><br>

        Salary:
        <input type="number" step="0.01" name="salary"
value="<%= rs.getDouble("salary") %>" required><br><br>

        <input type="submit" value="Update Employee">
```

```
    </form>
</body>
</html>
<%
    } else {
        out.println("Employee not found.");
    }
} catch (Exception e) {
    out.println("Error: " + e.getMessage());
} finally {
    if (rs != null) rs.close();
    if (ps != null) ps.close();
    if (con != null) con.close();
}
%>
```

update.jsp:

```
<%@ page import="java.sql.*" %>
<%
try {
    Class.forName("com.mysql.cj.jdbc.Driver");

    Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/EmployeeDB",
        "root",
        "admin@123"
    );

    PreparedStatement ps = con.prepareStatement(
        "UPDATE Employee SET name=?, department=?, salary=? 
WHERE emp_id=?"
    );

    ps.setString(1, request.getParameter("name"));
    ps.setString(2, request.getParameter("department"));
    ps.setDouble(3, 
Double.parseDouble(request.getParameter("salary")));
    ps.setInt(4, 
Integer.parseInt(request.getParameter("id")));

    ps.executeUpdate();

    ps.close();
    con.close();

    response.sendRedirect("index.jsp");
} catch (Exception e) {
```

```
    out.println("Error: " + e.getMessage());
}
%>
```

delete.jsp:

```
<%@ page import="java.sql.*" %>
<%
try {
    Class.forName("com.mysql.cj.jdbc.Driver");

    Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/EmployeeDB",
        "root",
        "admin@123"
    );

    PreparedStatement ps = con.prepareStatement(
        "DELETE FROM Employee WHERE emp_id=?"
    );

    ps.setInt(1,
Integer.parseInt(request.getParameter("id")));
    ps.executeUpdate();

    ps.close();
    con.close();

    response.sendRedirect("index.jsp");
} catch (Exception e) {
    out.println("Error: " + e.getMessage());
}
%>
```

Save.jsp :

```
<%@ page import="java.sql.*" %>
<%
try {
    Class.forName("com.mysql.cj.jdbc.Driver");

    Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/EmployeeDB",
        "root",
        "admin@123"
    );

    PreparedStatement ps = con.prepareStatement(
```

```
        "INSERT INTO Employee VALUES (?, ?, ?, ?)"
    );

    ps.setInt(1,
Integer.parseInt(request.getParameter("id")));
    ps.setString(2, request.getParameter("name"));
    ps.setString(3, request.getParameter("department"));
    ps.setDouble(4,
Double.parseDouble(request.getParameter("salary")));

    ps.executeUpdate();

    ps.close();
    con.close();

    response.sendRedirect("index.jsp");
} catch (Exception e) {
    out.println("Error: " + e.getMessage());
}
%>
```

Employee Management System

Employee ID:

Name:

Department:

Salary:

Add Employee

Employee Records

ID	Name	Department	Salary	Actions
123	piyush	CSE	123123.0	Edit | Delete

2. Create a JSP-based application to manage patient records using CRUD operations. Include fields such as patient ID, name, age, disease, and treatment status.

patient_index.jsp

```
<%@ page import="java.sql.*" %>
<html>
<head>
    <title>Patient Management System</title>
</head>
<body>
    <h2>Patient Management System</h2>

    <form action="save_patient.jsp" method="post">
        Patient ID: <input type="number" name="id"
required><br><br>
        Name: <input type="text" name="name" required><br><br>
        Age: <input type="number" name="age" min="0"
required><br><br>
        Disease: <input type="text" name="disease"
required><br><br>

        Treatment Status:
        <select name="status">
            <option value="Admitted">Admitted</option>
            <option value="Under Treatment">Under Treatment</
option>
            <option value="Recovered">Recovered</option>
            <option value="Discharged">Discharged</option>
        </select><br><br>

        <input type="submit" value="Add Patient">
    </form>

    <h3>Patient Records</h3>

    <table border="1" cellpadding="8">
        <tr>
            <th>Patient ID</th>
            <th>Name</th>
            <th>Age</th>
            <th>Disease</th>
            <th>Treatment Status</th>
            <th>Actions</th>
        </tr>

        <%
        try {
```

```
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/PatientDB",
                "root",
                "admin@123"
            );

            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery("SELECT * FROM
Patient");

            while (rs.next()) {
        %>
        <tr>
            <td><%= rs.getInt("patient_id") %></td>
            <td><%= rs.getString("name") %></td>
            <td><%= rs.getInt("age") %></td>
            <td><%= rs.getString("disease") %></td>
            <td><%= rs.getString("treatment_status") %></td>
            <td>
                <a href="edit_patient.jsp?id=<%=
rs.getInt("patient_id") %>">Edit</a>
                |
                <a href="delete_patient.jsp?id=<%=
rs.getInt("patient_id") %>" onclick="return confirm('Delete
this patient?')">Delete</a>
            </td>
        </tr>
        <%
            }

            rs.close();
            st.close();
            con.close();
        } catch (Exception e) {
            out.println("<tr><td colspan='6'>Error: " +
e.getMessage() + "</td></tr>");
        }
        %>
    </table>
</body>
</html>
```

save_patient.jsp:

```
<%@ page import="java.sql.*" %>
<%
try {
    Class.forName("com.mysql.cj.jdbc.Driver");

    Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/PatientDB",
        "root",
        "admin@123"
    );

    PreparedStatement ps = con.prepareStatement(
        "INSERT INTO Patient VALUES (?, ?, ?, ?, ?)"
    );

    ps.setInt(1,
Integer.parseInt(request.getParameter("id")));
    ps.setString(2, request.getParameter("name"));
    ps.setInt(3,
Integer.parseInt(request.getParameter("age")));
    ps.setString(4, request.getParameter("disease"));
    ps.setString(5, request.getParameter("status"));

    ps.executeUpdate();

    ps.close();
    con.close();

    response.sendRedirect("patient_index.jsp");
} catch (Exception e) {
    out.println("Error: " + e.getMessage());
}
%>
```

edit_patient.jsp:

```
<%@ page import="java.sql.*" %>
<%
Connection con = null;
PreparedStatement ps = null;
ResultSet rs = null;

try {
    Class.forName("com.mysql.cj.jdbc.Driver");

    con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/PatientDB",
        "root",
```

```
        "admin@123"
    );

    ps = con.prepareStatement(
        "SELECT * FROM Patient WHERE patient_id=?"
    );

    ps.setInt(1,
Integer.parseInt(request.getParameter("id")));
    rs = ps.executeQuery();

    if (rs.next()) {
        int id = rs.getInt("patient_id");
        String name = rs.getString("name");
        int age = rs.getInt("age");
        String disease = rs.getString("disease");
        String status = rs.getString("treatment_status");
%>
<html>
<head>
    <title>Edit Patient</title>
</head>
<body>
    <h2>Update Patient Details</h2>

    <form action="update_patient.jsp" method="post">
        Patient ID:
        <input type="number" name="id" value="<%= id %>"
readonly><br><br>

        Name:
        <input type="text" name="name" value="<%= name %>"
required><br><br>

        Age:
        <input type="number" name="age" value="<%= age %>"
min="0" required><br><br>

        Disease:
        <input type="text" name="disease" value="<%= disease
%>" required><br><br>

        Treatment Status:
        <select name="status">
            <option value="Admitted" <%=
status.equals("Admitted") ? "selected" : "" %>>Admitted</
option>
```

```
            <option value="Under Treatment" <%=
status.equals("Under Treatment") ? "selected" : "" %>>Under
Treatment</option>
            <option value="Recovered" <%=
status.equals("Recovered") ? "selected" : "" %>>Recovered</
option>
            <option value="Discharged" <%=
status.equals("Discharged") ? "selected" : "" %>>Discharged</
option>
        </select><br><br>

        <input type="submit" value="Update Patient">
    </form>
</body>
</html>
<%
    } else {
        out.println("Patient not found.");
    }
} catch (Exception e) {
    out.println("Error: " + e.getMessage());
} finally {
    if (rs != null) try { rs.close(); } catch (SQLException e)
{}
    if (ps != null) try { ps.close(); } catch (SQLException e)
{}
    if (con != null) try { con.close(); } catch (SQLException
e) {}
}
%>
```

Update_patient.jsp :

```
<%@ page import="java.sql.*" %>
<%
try {
    Class.forName("com.mysql.cj.jdbc.Driver");

    Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/PatientDB",
        "root",
        "admin@123"
    );

    PreparedStatement ps = con.prepareStatement(
        "UPDATE Patient SET name=?, age=?, disease=?,
treatment_status=? WHERE patient_id=?"
    );

    ps.setString(1, request.getParameter("name"));
```

```
    ps.setInt(2,
Integer.parseInt(request.getParameter("age")));
    ps.setString(3, request.getParameter("disease"));
    ps.setString(4, request.getParameter("status"));
    ps.setInt(5,
Integer.parseInt(request.getParameter("id")));

    int rows = ps.executeUpdate();

    ps.close();
    con.close();

    response.sendRedirect("patient_index.jsp");
} catch (Exception e) {
    out.println("Error: " + e.getMessage());
}
%>
```

delete_patient.jsp:

```
<%@ page import="java.sql.*" %>
<%
try {
    Class.forName("com.mysql.cj.jdbc.Driver");

    Connection con = DriverManager.getConnection(
        "jdbc:mysql://localhost:3306/PatientDB",
        "root",
        "admin@123"
    );

    PreparedStatement ps = con.prepareStatement(
        "DELETE FROM Patient WHERE patient_id=?"
    );

    ps.setInt(1,
Integer.parseInt(request.getParameter("id")));
    ps.executeUpdate();

    ps.close();
    con.close();

    response.sendRedirect("patient_index.jsp");
} catch (Exception e) {
    out.println("Error: " + e.getMessage());
}
%>
```

Patient Management System

Patient ID:

Name:

Age:

Disease:

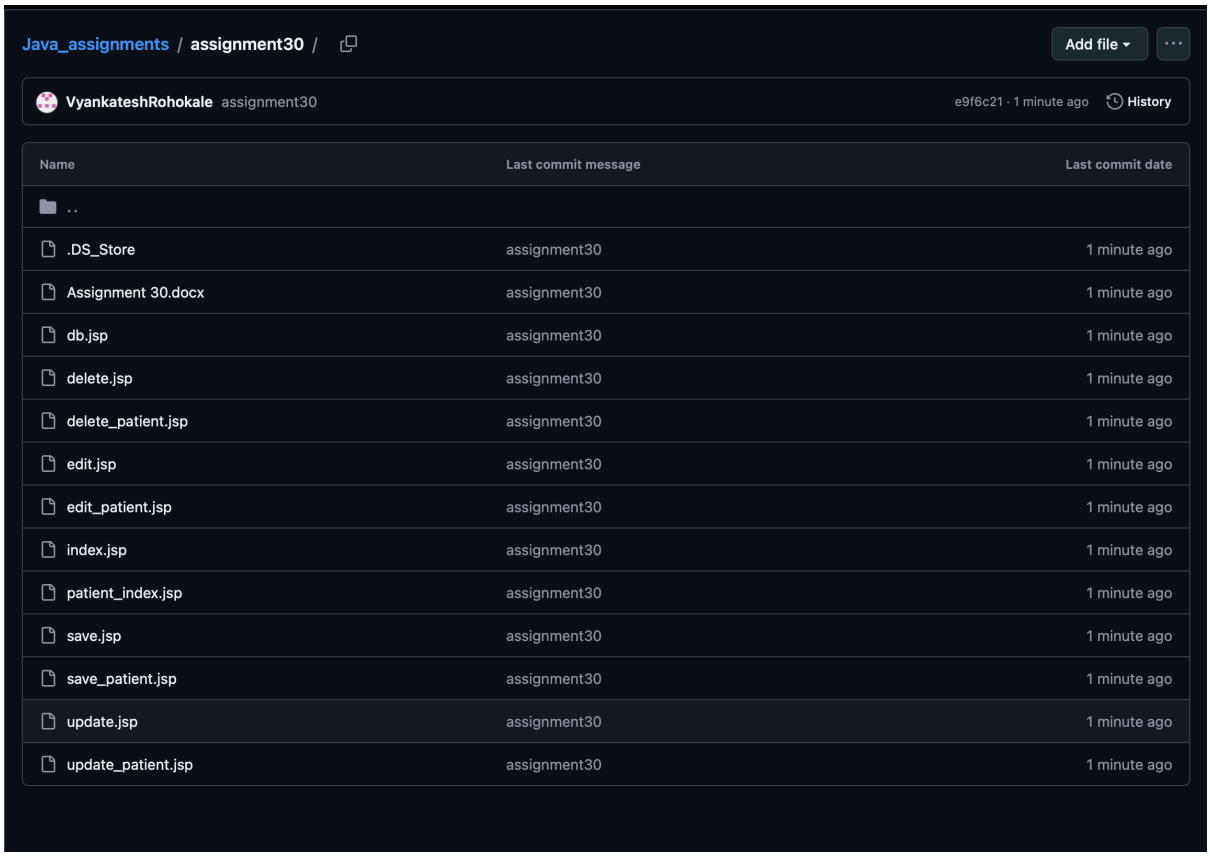
Treatment Status: Admitted

Add Patient

Patient Records

Patient ID	Name	Age	Disease	Treatment Status	Actions
123	Vyankatesh	19	mental illness	Under Treatment	Edit | Delete

Github:



Conclusion:

a JSP-based Employee Management System was developed using JSP, JDBC, and MySQL to perform CRUD operations, including adding, viewing, updating, and deleting employee records.